Learning Illumination for Depth-Resolved Fluorescence Microscopy

Anonymous CVPR submission

Abstract

Recovering fluorescent structures that overlap in projection requires measurements that distinguish their axial arrangements. We learn depth-encoding illumination for fixed-focus three-dimensional fluorescence microscopy using an angle-scanned plane wave and a fixed, defocused phase plate. The plate is optimised through the optical forward model by reducing correlations between detector-integrated voxel responses at different depths and the same lateral position. This objective compares finite depth changes across the angular acquisition, without specimen examples or a reconstruction decoder. We evaluate the learned plate with total variation and plug-and-play reconstruction using pretrained and neuron-trained denoisers. On simulated measurements of 60 mouse pyramidal neuron crops, it reduces axial-profile Wasserstein error by 28–50 % relative to its random initialisation at eightfold measurement compression and equal deposited excitation dose. Controlled crossing-filament tests also show improved separation under two reconstruction methods. Matched comparisons with worst-case coherence and a depth-Fisher surrogate favour the proposed criterion across both optimisation seeds and all three priors. The gains persist under background and read noise, and the learned plate achieves lower profile error at reduced dose. These results establish a depth-specific approach to optical learning whose benefits extend across the tested reconstruction priors.

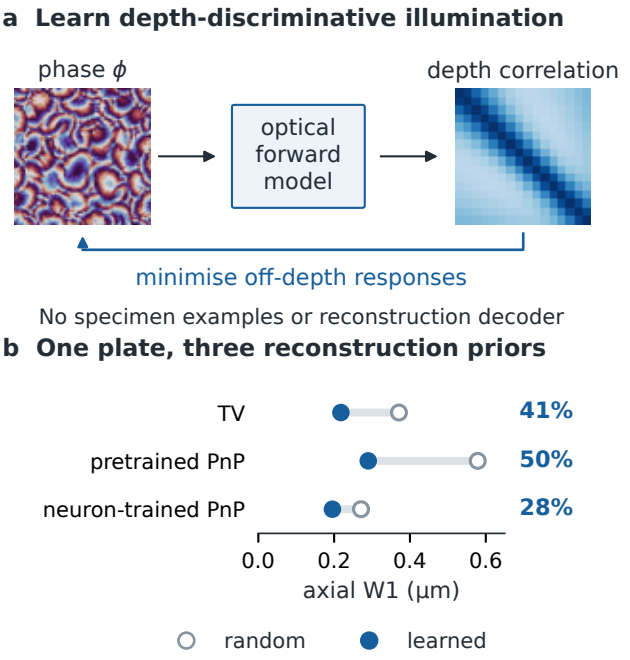


Figure 1. **Learn the illumination, then fix the plate for reconstruction.** (a) Optical training reduces correlations between depth-dependent measurement signatures. (b) The same learned plate improves axial-profile recovery with three reconstruction priors at eightfold measurement compression and equal deposited excitation dose. Means are over 60 neuronal crops; intervals are reported in Tab. 1.

1. Introduction

Recovering three-dimensional fluorescent structure from fixed-focus images requires measurements that distinguish different axial arrangements. With broad-depth-of-field detection, neuronal branches that overlap in projection can produce similar observations even when they lie at different depths. Reconstruction then relies more heavily on prior assumptions [2, 6]. We investigate how learning the excitation illumination can improve this discrimination before reconstruction.

Our system scans the incident angle of a plane wave through a fixed, defocused phase plate. Propagation produces spatially structured excitation, and changes in angle shift the patterns differently at different depths (Fig. 2). Structures at different axial positions therefore experience different sequences of excitation intensities. A fixed-focus camera records the resulting fluorescence images while the plate, specimen and detection focus remain stationary. Depth is encoded through the illumination sequence, without axial specimen scanning or detection refocusing.

The phase plate determines the measurement responses and is therefore central to this encoding. Previous work designs depth-sensitive optics through Fisher information and response coherence [20, 24], while information-driven

methods optimise optical measurements independently of a reconstruction decoder [13, 17]. We apply this measurement-driven principle to angle-scanned excitation. At each lateral position, we compare the detector-integrated responses of voxels at different depths and minimise their weighted squared correlations. These signatures include excitation modulation, emission blur and pixel averaging across angles. The objective penalises off-depth responses in a normalised signature comparison, providing a direct surrogate for finite-depth discrimination. Training requires no specimen examples, fitted measurement distribution or reconstruction solver.

We test whether the resulting optical design improves recovered depth profiles. Evaluation pairs the learned plate with its random initialisation under the same acquisition controls and uses total variation (TV), pretrained plug-and-play (PnP), and neuron-trained PnP. Simulated measurements of mouse pyramidal neuron volumes establish full-profile recovery, while controlled crossing filaments test separation and correct axial ordering. A separate comparison of expected photon counts tests distinguishability before reconstruction. We also compare optical objectives and examine measurement count, dose, noise and calibration.

Our contributions are:

1. **Fixed-focus excitation encoding.** We develop an angle-scanned fluorescence imaging system with a fixed, defocused phase plate and a voxel-integrated vectorial model of its depth-dependent measurements.
2. **Depth-signature optical learning.** We learn the plate by reducing weighted correlations between complete measurement responses at different depths, without specimen examples or a reconstruction decoder. Controlled comparisons evaluate this criterion against coherence and Fisher-inspired objectives.
3. **Recovery across priors and acquisition budgets.** We demonstrate improved axial-profile recovery under three priors, smaller tested crossing separations under two of them, and useful measurement-count and excitation-dose trade-offs.

2. Related work

Depth encoding in fluorescence microscopy. Three-dimensional fluorescence microscopy can encode depth through detection or excitation. Light-field microscopy, DiffuserCam and Miniscope3D recover volumes from depth-dependent detection responses [1, 4, 24]. Excitation-based methods instead use illumination patterns that vary with depth. Yeh *et al*. [25] combine translated speckle patterns with axial scanning, while Pascucci *et al*. [15] exploit axial speckle decorrelation and fluorescence saturation for volumetric recovery from a two-dimensional scan. Related approaches use angular scanning through a fixed scattering layer for two-dimensional fluorescence recovery [26], or tilted pattern projection to produce depth-dependent lateral shifts [8]. We study linear fluorescence with angular scanning through a fixed, defocused phase plate, and learn its phase profile to improve axial discrimination.

Learning optical encoders. Optical design can be guided by reconstruction performance or by properties of the measurements. End-to-end methods jointly optimise optics and reconstruction networks [7, 14, 21], while Kellman *et al*. [10] optimise illumination for a specified reconstruction algorithm. Decoder-independent methods instead optimise optical responses directly. Shechtman *et al*. [20] use Fisher information for 3D localisation, while Miniscope3D [24] optimises a detection-side phase mask using cross-coherence between depth-dependent point-spread functions, including lateral shifts. Similar coherence objectives appear in sensing-matrix design [5]. IDEAL and IDEAL-IO [13, 17] optimise estimated mutual information using fitted measurement-distribution models, without a reconstruction decoder. Our excitation-side objective follows this measurement-driven principle. It compares detector-integrated voxel responses across illumination angles, reducing weighted squared correlations between depths at each lateral position. Training needs no specimen examples, fitted measurement distribution or reconstruction solver. We evaluate the resulting design through axial-profile recovery and controlled separation tests.

Volumetric reconstruction and priors. TV regularisation promotes piecewise-smooth structure [19], while PnP incorporates denoisers as implicit priors [9, 22]. Multi-slice fusion applies lower-dimensional denoisers along complementary orientations of a shared volume [11]. We use these established approaches to evaluate the optical design under different prior assumptions, keeping the priors fixed across encoders. Measurement-level comparisons and controlled axial-separation tests complement reconstruction accuracy.

3. Learning depth-encoding illumination

Acquisition. For transverse position $\mathbf{r} = (x, y)$, illumination direction t produces the incident plane wave $u_t^{\mathrm{in}}(\mathbf{r}) = a_t e^{\mathrm{i}k_{\mathrm{ex}}\mathbf{s}_t^\top \mathbf{r}}$, where $\mathbf{s}_t$ is the transverse direction-cosine vector and $k_{\mathrm{ex}} = 2\pi/\lambda_{\mathrm{ex}}$ for refractive index one. The angular grid is specified in the supplement. The field passes through the fixed phase plate, $\tau(\mathbf{r}) = e^{\mathrm{i}\phi(\mathbf{r})}$, and propagates a distance D to the intermediate illumination image plane by angular-spectrum propagation. The objective pupil filters the modulated field before a Richards–Wolf relay [12, 18] produces the specimen excitation. We use a vectorial model for the high-NA relay, with x-polarised input and intensity $I_t = \sum_{c\in\{x,y,z\}} |E_{t,c}|^2$. Coordinates are in specimen-equivalent units. One fluorescence image is recorded per

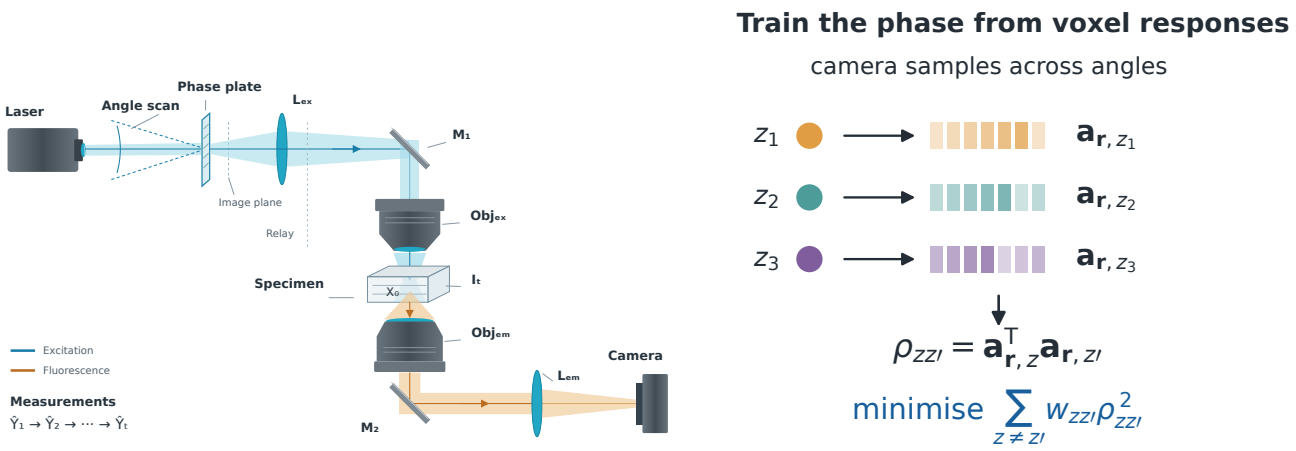


Figure 2. **Fixed-focus acquisition and depth-signature learning.** Left: only the incident angle changes; the phase plate, specimen and detection focus remain stationary. Blue: excitation; orange: fluorescence. Right: schematic assembly of voxel responses across camera samples and illumination angles. Training compares depths at the same lateral position and updates the phase through the optical model. The specimen and a reconstruction decoder are not used in optical training.

direction at a fixed detection focus; acquisition is sequential.

Depth encoding. Propagation after the plate creates spatial intensity structure. Changing the incident angle shifts this structure laterally, with displacement increasing with propagation distance. In the paraxial thin-screen approximation, a change $\Delta\mathbf{s}$ produces $\Delta\mathbf{r} \approx d\,\Delta\mathbf{s}$ after distance d. Different depths therefore experience different excitation sequences during the scan. Plate defocus places the shear pivot outside the specimen. This relation supplies intuition; the vectorial simulation includes finite-pupil effects and departures from translation.

Image formation. Let $X_0(\mathbf{r}, z)$ denote fluorescence response per unit excitation. Under linear, unsaturated fluorescence, the emission source is proportional to $X_0 I_t$ [25]. For laterally shift-invariant detection, the noiseless camera-plane irradiance is

$$g_t(\mathbf{r}) = \int_\Omega h_{\mathrm{em}}(\mathbf{r}-\mathbf{r}', z_{\mathrm{focal}}-z')X_0(\mathbf{r}', z')I_t(\mathbf{r}', z')\,\mathrm{d}^2\mathbf{r}'\,\mathrm{d}z', \tag{1}$$

where h_{em} is the emission intensity PSF and Ω the specimen volume. Camera pixel p, occupying D_p, records $\hat{Y}^0_{t,p} = |D_p|^{-1}\int_{D_p} g_t(\mathbf{r})\,\mathrm{d}^2\mathbf{r}$. Fluorescence contributions add incoherently before pixel averaging. Excitation mixes specimen spatial frequencies before detection blur, allowing the angle sequence to carry information that a single image does not resolve.

Voxel-integrated responses. We evaluate the optics on a finer grid and integrate responses within each reconstruction voxel. If B is the fine-grid operator and U replicates reconstruction coefficients over their fine-grid cells, the reconstruction operator is $\mathsf{A} = \mathsf{BU}$. This is the exact restriction of the discretised model to a piecewise-constant basis. Sub-voxel specimen variation remains an approximation error. The assembly formula, residual model and computational details are given in the supplement.

Learning the plate. For a sampled angle set $\mathcal{T}$, a voxel's restricted operator column collects its camera response across those angles. At lateral position $\mathbf{r}$ and depth z, let $\mathbf{a}_{\mathbf{r},z}$ be this column after centring and unit-norm normalisation. We optimise the phase by minimising

$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{r},\mathcal{T}}\frac{\sum_{z\neq z'} w_{zz'}\left(\mathbf{a}_{\mathbf{r},z}^\top \mathbf{a}_{\mathbf{r},z'}\right)^2}{\sum_{z\neq z'} w_{zz'}}. \tag{2}$$

Selected symmetric depth pairs receive weight two; other pairs receive weight one. Squaring penalises correlations of either sign, while normalisation prevents uniform amplification from reducing the loss. Gradients pass through excitation, vectorial propagation, emission blur and pixel averaging. No specimen examples, density model or reconstruction solver enters training.

Interpretation. Stack the normalised signatures as columns of $\mathsf{S}_{\mathbf{r}}$ and write $\mathsf{G}_{\mathbf{r}} = \mathsf{S}_{\mathbf{r}}^\top\mathsf{S}_{\mathbf{r}}$. Comparing one signature with this bank gives its column of $\mathsf{G}_{\mathbf{r}}$: the diagonal response is one, and off-diagonal entries are responses at other depths. Apart from constant normalisation, the unweighted loss is $\|\mathsf{G}_{\mathbf{r}} - \mathsf{I}\|_F^2$. The weights emphasise selected axial alternatives. This explains what the criterion reduces in signature space; reconstruction uses the original operator A.

4. Experimental setup

All experiments use the simulated fixed-focus fluorescence microscope described in Sec. 3. We use a common-wavelength approximation of 532 nm for excitation and emission, with numerical apertures of 0.8 and 0.1, respectively. The phase plate has $0.32\,\mu\mathrm{m}$ features and is positioned $6.6\,\mu\mathrm{m}$ before the intermediate illumination image plane, in specimen-equivalent coordinates. We use 256 illumination directions. Optical fields are evaluated with $0.08\,\mu\mathrm{m}$ lateral and $0.32\,\mu\mathrm{m}$ axial sampling, followed by $0.32\,\mu\mathrm{m}$ analysis sampling and $0.64\,\mu\mathrm{m}$ reconstruction voxels. The model assumes linear fluorescence in a homogeneous, non-scattering medium.

Volumes and compression. We reconstruct volumes on a $16 \times 48 \times 48$ voxel grid ($N = 36{,}864$), covering $30.7 \times 30.7 \times 10.2\,\mu\mathrm{m}^3$ (lateral $\times$ lateral $\times$ axial). Each camera image contains 12×12 samples. Using $k = 256$, 64, or 32 illumination directions gives measurement compression factors $C = N/(144k)$ of 1, 4, and 8, respectively. For $C > 1$, we evaluate five randomly selected direction subsets. Reference volumes are obtained from two-photon stacks of mouse pyramidal neurons in the public DEEP^2 dataset [23], min–max normalised and resampled to the analysis grid.

Dataset splits and validation. The test set contains six neuronal cells from three animals, with two prespecified crops per cell selected by a structure score. With reconstruction parameters fixed, we randomly selected eight additional crops per cell, giving 60 test crops for the cross-decoder evaluation (Sec. 5.2). Unless otherwise stated, other full-field experiments use the 12 prespecified crops. Neuron-trained denoisers are trained on 459 crops from a fourth animal. Reconstruction parameters are selected separately for each encoder and compression setting using two vascular crops from a separate stack acquired from a different animal. No test volumes are used for training, tuning, or calibration.

Reconstruction priors. We evaluate three methods, none trained jointly with the plate. *TV* uses box-constrained least squares with isotropic three-dimensional total variation [19], solved by ADMM [3]. *Pretrained PnP* uses a natural-image DnCNN [27] along the xy, yz and zx slice orientations of a shared volume, with a sparsity branch, following multi-slice fusion [11]. *Neuron-trained PnP* replaces the slice denoiser with a neuron-trained 2D network and adds a neuron-trained 3D branch. Both PnP configurations combine measurement consistency, denoising and sparsity through PnP-ADMM [9, 22]. Priors remain fixed across encoders. Solver updates, denoiser training and settings are detailed in the supplement.

Optical controls and dose. The main comparison pairs the learned plate with its random initialisation. Two optimisation seeds change the sampled positions and directions while retaining this initial phase. Additional random plates, a defocused amplitude mask, a plate-free scan and an in-focus DMD are evaluated as supporting controls. Excitation is simulated beyond the specimen field and normalised to a common mean over the full scan and specimen volume. For k acquired directions, counts follow $\mathrm{Poisson}(\kappa_0\, 256\, \bar{y}_i/k)$, where $\bar{y}_i$ is the noiseless measurement. The factor $256/k$ maintains nominal deposited dose by increasing exposure per image. For each specimen, κ_0 gives the random plate 1000 expected photons per pixel per image at the mean signal for $k = 256$ and is shared across encoders.

Axial-profile and image-quality metrics. Our primary depth metric is the axial Wasserstein-1 distance [16]. At each lateral column c, we normalise the non-negative reference and reconstructed axial profiles and form their cumulative distributions F_c and $\hat{F}_c$. The reported error is

$$E_{\mathrm{W1}} = \frac{\Delta z}{\sum_{c\in\mathcal{S}} w_c} \sum_{c\in\mathcal{S}} w_c \sum_{\ell} \left| F_c(z_\ell) - \hat{F}_c(z_\ell) \right|, \qquad (3)$$

where $\Delta z = 0.64\,\mu\mathrm{m}$, w_c is the reference column brightness, and $\mathcal{S}$ contains columns exceeding 10 % of the maximum reference column brightness. This measures axial-profile displacement in micrometres and detects merging that can preserve the centroid. We also report the signal-weighted absolute centroid error over the same columns, PSNR and foreground SSIM. W1 normalises column brightness; the image-quality metrics retain sensitivity to intensity and lateral errors. Depth-blind references and additional metrics are detailed in the supplement.

Measurement distinguishability. For each reference we construct a merged-depth alternative with the same lateral projection and column centroids. We compare their expected fine-grid photon measurements using Poisson KL, fitting the alternative's unknown global brightness. This decoder-free test uses the 59 crops containing layered columns. Exact alternatives, nuisance fitting and background variants are specified in the supplement.

Crossing structures. Success requires separating both depths at a filament crossing and correctly ordering the arms. The threshold is the smallest tested gap with at least 80 % success at each of two axial centres and all larger tested gaps, using three geometries and five draws. A merged control tests false splitting. Peak, ordering and phantom definitions are given in the supplement.

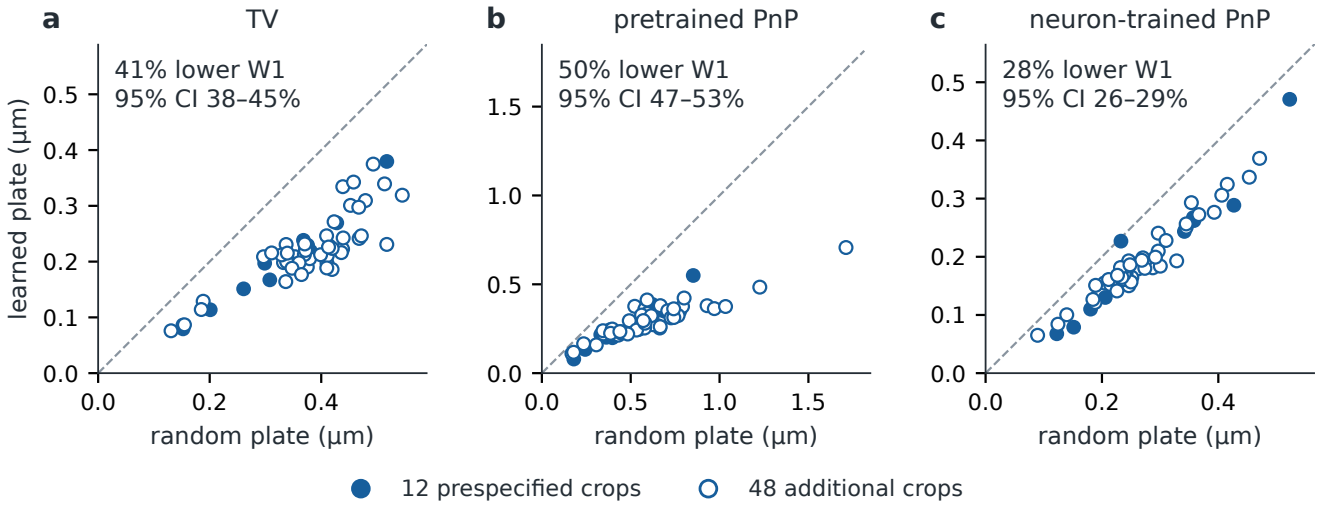


Figure 3. **Axial-profile recovery improves under each prior.** Each point is one neuronal crop, averaged over five measurement draws at $C = 8$ and equal dose. The dashed line denotes equal W1. Filled and open markers distinguish the prespecified and additional crops. All 60 crops improve under each method. Percentage reductions compare paired means; 95 % intervals resample cells. Axes use equal horizontal and vertical scales within each panel.

	axial W1 (μm) ↓		centroid error (μm) ↓		PSNR (dB) ↑		W1 reduction
prior	random	learned	random	learned	random	learned	[95 % CI]
TV	0.371	0.218	0.259	0.150	36.70	38.29	41 % [38, 45]
pretrained PnP	0.579	0.290	0.490	0.224	35.72	37.56	50 % [47, 53]
neuron-trained PnP	0.271	0.195	0.197	0.139	38.01	39.60	28 % [26, 29]

Table 1. **Effect of optical learning with fixed reconstruction priors.** Mean over 60 neuronal crops at eightfold measurement compression and equal deposited excitation dose. W1, centroid error and PSNR use the same reconstructions. Draws are averaged within each crop before averaging crops; W1-reduction intervals use paired cell-bootstrap resampling.

Statistics. Comparisons pair specimens, angular subsets, dose and noise protocol, averaging draws within each crop. Intervals use 10,000 bootstrap resamples of the six cells with their crops retained together. Error reductions compare paired means; KL ratios are summarised geometrically.

5. Results

5.1. Learning depth-discriminative responses

Learning reorganises the phase plate and reduces mean absolute depth-signature correlation from 0.84 to 0.48. Before/after maps and the measurement-comparison plot are given in the supplement. The measurement-level comparison tests a consequence beyond this training statistic. Its alternative preserves the lateral projection and column centroids, so the test concerns a change in axial arrangement that those summaries do not capture. Relative to the random plate, profiled Poisson KL to the merged-depth alternative rises by a geometric mean factor of 10.0 (95 % CI: 8.7–11.8), increasing in all 59 eligible neuronal crops. With added background, the factor is 10.4. Thus, the learned measurements distinguish these axial alternatives more strongly before any reconstruction prior is applied. This diagnostic compares selected alternatives; it does not measure global inverse stability.

5.2. Recovering neuronal volumes across priors

At $C = 8$, the learned plate reduces mean axial W1 by 41 % with TV, 50 % with pretrained PnP and 28 % with neuron-trained PnP (Fig. 3 and Tab. 1). All cell-bootstrap intervals exclude zero, and every crop improves under each prior. The 48 randomly selected crops were added after reconstruction parameters were fixed, supporting the result beyond the 12 prespecified crops. A second optimisation seed gives similar reductions of 41 %, 50 % and 29 % from the same initial phase.

Centroid error falls by 30–54 %, showing improved axial placement alongside the profile gains, and PSNR increases by 1.6–1.8 dB (Tab. 1). On reference columns containing multiple axial peaks, TV reduces W1 by 34 %. Representa-

prior	random $1\times$ dose	learned $0.3\times$ dose	reduction [95 % CI]
TV	0.318	0.234	26 % [20, 33]
pretrained PnP	0.496	0.302	39 % [34, 45]
neuron-trained PnP	0.282	0.220	22 % [16, 28]

Table 2. **Recovery at reduced excitation dose.** Mean W1 (μm, $C = 8$, 12 prespecified crops). Reductions compare learned at $0.3\times$ with random at $1\times$ dose, with cell-bootstrap 95 % intervals.

tive volume views, column profiles and the complete image-quality results are provided in the supplement.

The optical and prior gains are complementary in these comparisons. Neuron-trained PnP has the lowest random-plate W1, yet still benefits from learning the illumination. Conversely, the learned plate with TV has lower mean W1 than the random plate with neuron-trained PnP (0.218 versus $0.271\,\mu$m). Thus, improved measurements can benefit recovery even when the reconstruction uses a conventional prior; a stronger learned prior does not remove the optical design opportunity.

5.3. Comparing optical training objectives

The proposed criterion compares finite depth alternatives through their complete recorded responses. We compare it with worst-case coherence and a depth-Fisher surrogate under matched initialisation, sampling and optimisation budgets (Fig. 4). All three objectives improve upon the random plate. Depth decorrelation gives 13–20 % lower W1 than the two alternatives across both optimisation seeds and all three priors. An uncentred ablation performs similarly to the centred objective under neuron-trained PnP. Full losses, absolute errors and the $C = 1$ comparison are given in the supplement.

5.4. Separating crossing structures

The crossing test checks whether better depth responses yield distinct recovered structures (Fig. 4). With TV and pretrained PnP, the smallest tested gap meeting the separation and ordering criterion falls from 3.84 to $2.56\,\mu$m. This holds at both axial centres, including the centre outside the mirror pairing emphasised during training. Neuron-trained PnP retains a $3.84\,\mu$m threshold, although success rates increase at smaller gaps. Profile accuracy and threshold separation need not coincide: neuron-trained PnP improves W1 by 28 % without changing this threshold. Neither plate falsely splits the merged control. Unequal-intensity and three-filament cases are reported in the supplement.

5.5. Measurement and dose trade-offs

At 0.3 times the reference excitation dose, the learned plate achieves lower W1 than the random plate at full dose under all three priors: reductions are 26 % [20, 33], 39 % [34, 45] and 22 % [16, 28] for TV, pretrained PnP and neuron-trained PnP (Tab. 2). At equal total dose, 32 learned-plate images also outperform 256 random-plate images under TV and pretrained PnP, while neuron-trained PnP gives similar means. Fewer images receive longer exposures in this comparison. Complete dose and compression curves are given in the supplement.

5.6. Noise and optical calibration

With background and read noise, W1 reductions remain 30 %, 53 % and 27 % under the three priors. Gains also persist when both encoders receive one tenth of the reference dose. For neuron-trained PnP, plate-distance offsets of $\pm 0.5\,\mu$m retain reductions of 27–28 % after updating the forward model. Uncalibrated pointing errors of 0.005–0.01 in direction cosine retain 13–20 % reductions. Uncalibrated axial or lateral plate offsets remove the advantage; a one-feature lateral shift makes W1 11 % worse than random (95 % CI: 2–23 %). The results therefore depend on optical registration as well as encoder design.

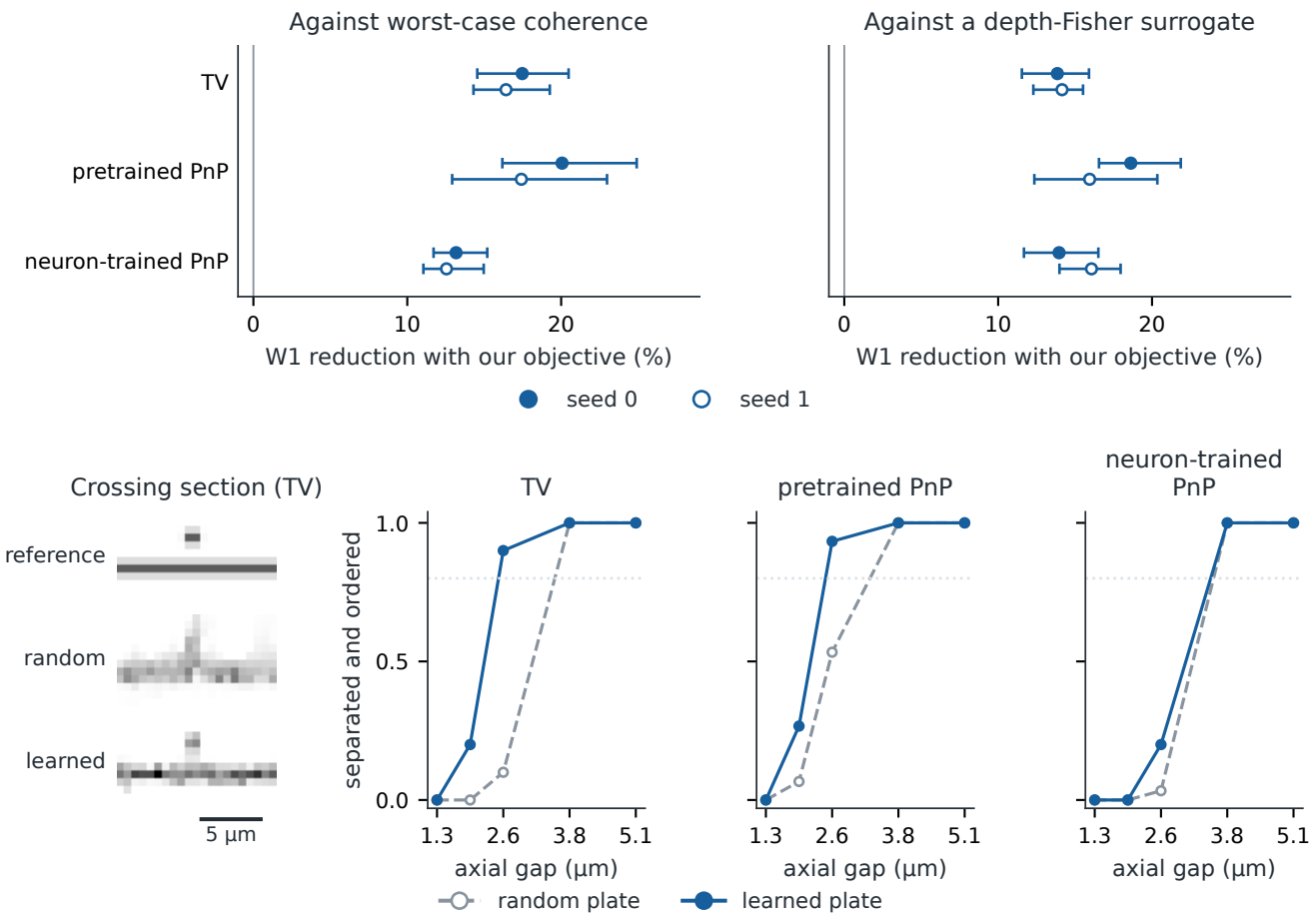


Figure 4. **Objective choice and axial separation.** Top: paired W1 reductions against the implemented coherence and depth-Fisher surrogate ($C = 8$, 12 crops), with matched training budgets and cell-bootstrap intervals; seeds change sampling. Bottom: TV crossing sections on a common intensity scale (bar: $5\,\mu$m) and success rates pooled across geometries, centres and draws. The 80 % threshold is checked at each centre and larger tested gaps. TV and pretrained PnP improve from 3.84 to $2.56\,\mu$m; neuron-trained PnP retains $3.84\,\mu$m.

6. Discussion and limitations

Optical design across priors. The learned plate improves axial profiles under both conventional and learned regularisation. Its gain under TV shows that the result does not depend on a neuron-trained denoiser. The two PnP methods share a framework, but differ in their training distribution and volumetric prior. Together with the measurement comparison, these results support designing depth responses independently of reconstruction. They do not compare this approach with end-to-end optical learning or establish transfer to arbitrary decoders.

Profile accuracy and separation. The loss targets finite axial changes at the same lateral position. It does not directly penalise confusion between laterally displaced voxels. Full-volume recovery nevertheless improves under all tested priors, while crossing thresholds improve under two of them. Lower profile error and smaller tested separation are complementary outcomes; neither establishes that every branch or connection is uniquely determined. The supplementary structural analyses preserve this distinction.

Physical scope. The study simulates linear fluorescence in a homogeneous, non-scattering medium, using broad-depth-of-field detection and idealised components. Aberrations, scattering and experimental imperfections can change the learned responses. The alignment results show that calibration is important: an updated model retains gains under moderate displacement, while uncalibrated offsets can remove them. A realised instrument therefore needs its own optical characterisation and validation.

Evidence scope. The main comparison uses 60 crops from six cells across three animals; cell-bootstrap intervals are approximate with six cells. Objective, dose and tolerance

comparisons use the 12 prespecified crops. The two optical optimisation seeds share one initial phase, and the neuron-trained denoisers use a fourth animal. The evidence supports the tested acquisition and specimen regime, with broader optical and specimen validation remaining open.

7. Conclusion

We learn fixed-focus fluorescence illumination by optimising depth-dependent measurement signatures through the optical model. Training uses no specimen examples or reconstruction decoder. The learned plate reduces axial-profile error by 28–50 % across TV and two PnP methods, improves tested crossing separations under two priors, and achieves lower profile error at 0.3 times the random plate's excitation dose. These simulation results support depth-signature design as a route to improved volumetric recovery across reconstruction priors.

LLM Usage Statement

LLM tools assisted with manuscript drafting, editing and figure-generation code.

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